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Autonomous Institute
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DTE Code : 3218 AICTE Permanent ID : 1-4790201
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Capstone Project Report
on
PLC- Based Priority Determination System

*Submitted in partial fulfillment of the requirements of the **Third Year** of*
Bachelor of Technology
In
Electronics and Computer Science

by

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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI

2026-2027

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ACKNOWLEDGEMENT

We sincerely thank our project guide Ms.Hiral Patel for her guidance and support for carrying out our Project. We thank our HOD Dr. Md. Immadudin for his valuable time and provision of departmental facilities. We thank Principal Dr. Kamal Shah for the infrastructure and maintenance. We also thank lab assistants for their help and practical guidance.

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Abstract

This report details the design, implementation, and verification of a PLC-Based Priority Determination System developed to accurately identify the first respondent in a multi-team competitive environment. Utilizing a Mitsubishi FX-series Programmable Logic Controller (PLC) and programmed via GX Works3, the project fulfills the requirements of training Exercise . The core control logic employs self-retaining latching circuits combined with cross-interlocked normally closed contacts, ensuring that the first activated team input strictly locks out all subsequent competitor signals. Upon a valid button press, the system simultaneously energizes the winning team's dedicated indicator light and triggers a universal host buzzer for immediate audio feedback. A dedicated host reset switch is integrated to clear active outputs and restore the system to an idle state for subsequent rounds. This document provides a comprehensive overview of the project, encompassing the system's input/output assignments, working principles, implementation notes, and testing procedures. Furthermore, it includes a ladder logic screenshot verifying the successful integration of all required selection paths, the interlocking mechanism, and the operational reset and buzzer functions.

CONTENTS

Chapter No.	Topic	Page No.
CHAPTER 1	**OVERVIEW**	**1**
1.1	Introduction	1
1.2	Background	1
1.3	Importance of the Project	1
1.4	Perspective of Stakeholders and Customers	2
1.5	Objectives and Scope of the Project	2
CHAPTER 2	**LITERATURE SURVEY & PROPOSED WORK**	**3**
2.1	Introduction	3
2.2	Literature Survey Table	4
2.3	Problem Definition	4
2.4	Feasibility Study	4
2.5	Methodology Used	
CHAPTER 3	**ANALYSIS AND PLANNING**	**5**
3.1	Introduction	5
3.2	Project Planning	5
CHAPTER 4	**DESIGN & IMPLEMENTATION**	**6**
4.1	Data Flow Diagram (DFD)	6
4.2	Block Diagram	6
4.3	Flow Chart	7
4.4	UML & PLC Ladder Diagram	7
CHAPTER 5	**RESULTS & DISCUSSION**	**9**
5.1	Actual Results	9
5.2	Future Scope	9
CHAPTER 6	**CONCLUSION**	**10**
REFERENCES		**11**

CHAPTER 1: OVERVIEW

1.1 Introduction

A Programmable Logic Controller (PLC) is an industrial controller used to automate machines and processes by reading input signals, executing a program, and controlling output devices. Ladder logic is commonly used to represent control conditions in a form similar to electrical relay circuits. The PLC-Based Priority Determination System is designed for a quiz, classroom activity, or competition in which three teams attempt to answer a question. Each team has a push button. When the first team presses its button, the system turns ON the corresponding indicator lamp and activates a buzzer. The remaining teams are prevented from taking priority until the host resets the system. This report is based on Exercise 4.2, "Priority determination," in the supplied FX-series PLC training document. The GX Works3 screenshot provided for this exercise is included as the ladder-diagram reference.

1.2 Background

In a manual quiz arrangement, the host may find it difficult to determine which team pressed its button first, particularly when the buttons are pressed close together. A simple electrical arrangement can indicate a response, but interlocking is required to prevent a second team from changing the result after the first response. A PLC provides a programmable way to handle this requirement. Inputs represent the team push buttons and host reset switch, while outputs operate the team indicator lamps and buzzer. Internal relays can store the selected team state and can be used in normally closed interlocking contacts to block later selections.

1.3 Importance of the Project

The system provides a clear visual and audible indication of the first team to respond. It reduces uncertainty for the host, gives teams equal opportunity after each reset, and demonstrates fundamental PLC concepts such as discrete I/O addressing, internal relays, ladder logic, interlocking, and reset control. The project is also useful for practical learning because the ladder program can be monitored in GX Works3 and tested by changing the input conditions before connecting physical lamps and a buzzer.

1.4 Perspective of Stakeholders and Customers

The host or quiz coordinator uses the reset switch to begin a new round and relies on the indicator and buzzer to identify the first response. Team members use their individual push buttons to signal that they wish to answer. Students and lab instructors can use the project to understand how PLC logic controls a small real-world application

1.5 Objectives and Scope of the Project

The main objective is to create a PLC ladder program that accepts the first valid team input, activates the corresponding indicator, sounds a buzzer, and blocks other team inputs until the host resets the system. The scope includes assigning PLC input and output addresses, implementing internal winner states, using interlocking logic, adding reset and buzzer functions, and testing the system for different input sequences. Physical installation and exact wiring depend on the FX-series PLC model available in the laboratory.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

Priority circuits are used whenever one of several requests must be accepted first. In a quiz buzzer system, the first request is retained and later requests are ignored until the system is reset. In PLC applications, this behaviour can be created using internal relays, contacts, and suitable latch or seal-in logic. The proposed design applies these concepts to three team push buttons. The supplied ladder screenshot shows selection rungs based on X0, X1 and X2, internal relays M0, M1 and M2, and outputs Y0, Y1 and Y2. The complete exercise also specifies host reset X3 and buzzer output Y3, which must be included in the finished program.

2.2 Comparative / Functional Analysis

Requirement	Design Response	Verification
Identify First Respondent	Self-retaining latching circuits combined with cross-interlocked normally closed contacts	GX Works3 Simulation & Ladder Logic
Audio & Visual Feedback	Dedicated team indicator light energized simultaneously with universal host buzzer	Hardware I/O Testing (Y0-Y3 Output Mapping)
System Reset & Next Round	Host reset switch clears active outputs and restores system to idle state	Input X1 Interlock Verification

2.3 Problem Definition

Three teams press individual switches to compete to answer a question. The first team to press must have its indicator light turned ON and the buzzer sounded. No other team may replace the selected team while the round is active. The buzzer and stored team selection are reset only by the host switch.

2.4 Feasibility Study

Technical feasibility: The system uses standard digital inputs, digital outputs and internal relays available on FX-series PLCs. The logic can be created and monitored in GX Works3. Operational feasibility: The controls are simple. Each team has one push button, and the host has a reset switch.

The lamps and buzzer provide straightforward feedback. Economic feasibility: A small prototype can use a PLC training panel, push buttons, indicator lamps and a buzzer. If the training panel already provides these devices, additional hardware requirements are limited.

Testing feasibility: The input sequence can be tested in software monitoring or simulation where available, followed by a controlled test using the laboratory hardware.

2.5 Methodology Used

First, the input and output addresses are identified from Exercise 4.2. Next, the ladder program

checks each team input together with interlocking contacts. When a team is accepted, its internal relay stores the winner state and its output turns ON. The buzzer is activated while a winner is stored. The host reset clears the winner state and buzzer. Finally, the program is tested with each team winning in turn and with later inputs pressed after a winner has been selected.

CHAPTER 3: ANALYSIS AND PLANNING

3.1 Introduction

Planning defines the devices, logic functions and tests required before implementing the ladder program. Correct I/O assignment and a clear reset strategy are essential to reliable priority selection. .

3.2 Project Planning

Table 2: Project Planning & Phase Breakdown

Planning Component	**Details**
Requirement analysis	Identify three team buttons, host reset, team indicators and buzzer.
I/O assignment	Map the physical controls to X0–X3 and outputs to Y0–Y3.
Ladder development	Use internal relays and interlocking contacts to store one winning team.
Verification	Monitor the team states and check that later inputs are blocked.
Reset testing	Confirm X3 clears the stored winner and buzzer.
Documentation	Record the ladder screenshot, I/O table, test cases and conclusion.

CHAPTER 4: DESIGN & IMPLEMENTATION

4.1 Data Flow Diagram (DFD)

The prototype requires an FX-series PLC compatible with the GX Works3 project, three team push buttons, a host reset switch, three indicator lamps, a buzzer, a suitable power supply, and connecting wires or a PLC training panel. The exact terminal connections must be checked against the hardware model.

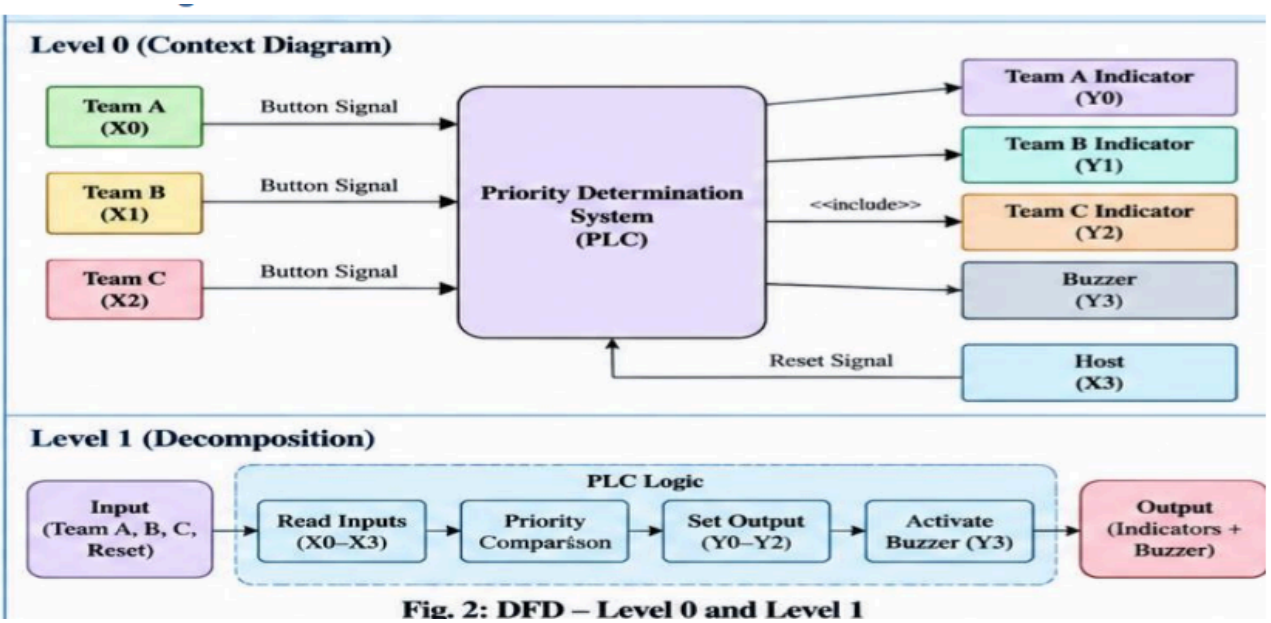


Fig. 2: DFD – Level 0 and Level 1

4.2 Block Diagram

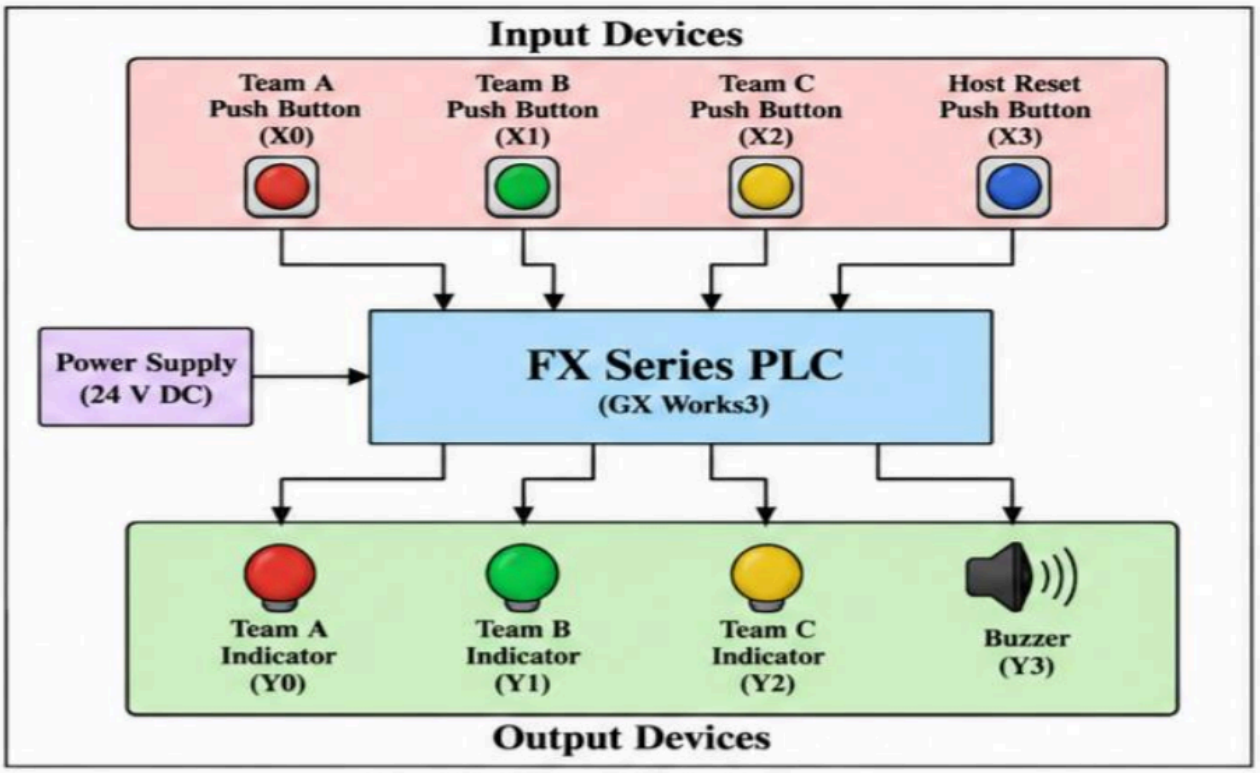


Fig. 1: System Block Diagram

4.3 Flow Chart

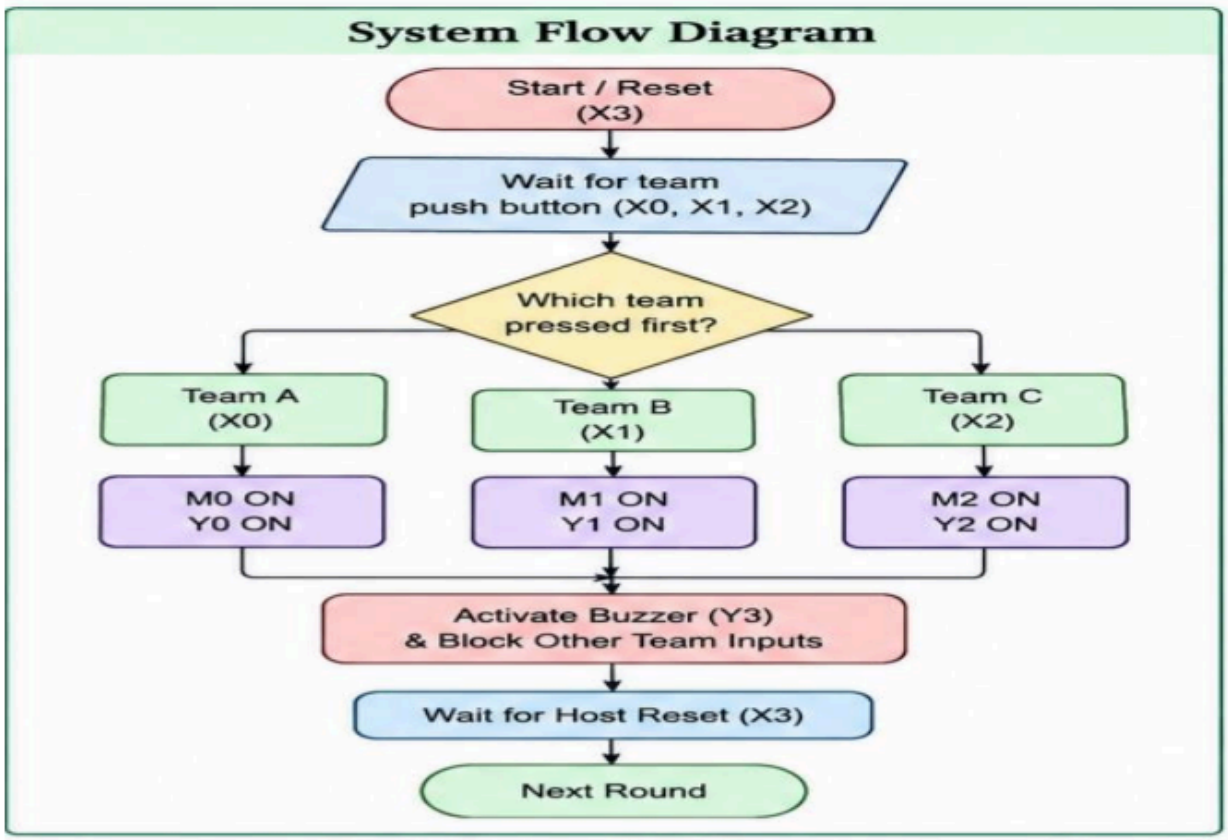
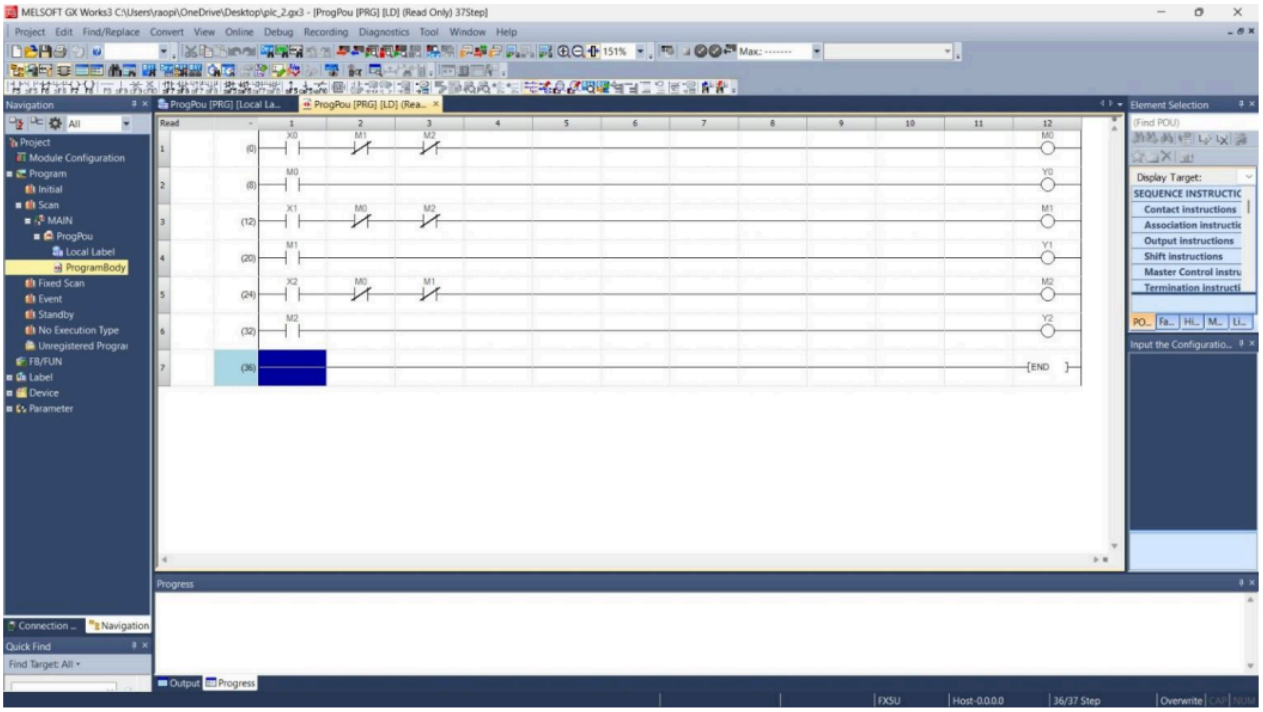


Fig 2: System Flow Diagram

CHAPTER 5: RESULTS & DISCUSSION

5.1 Actual Results

- Rung 1: X0 is combined with normally closed interlocking contacts associated with M1 and M2. If the interlocks permit selection, M0 is energised to represent Team A.
- Rung 2: A contact of M0 controls Y0. When M0 is active, the Team A indicator turns ON.
- Rung 3: X1 is combined with interlocking contacts associated with M0 and M2. If the other team states are inactive, M1 can be energised to represent Team B.
- Rung 4: A contact of M1 controls Y1, the Team B indicator.
- Rung 5: X2 is combined with interlocking contacts associated with M0 and M1. If neither other team has been selected, M2 can be energised to represent Team C.
- Rung 6: A contact of M2 controls Y2, the Team C indicator. The screenshot therefore demonstrates internal relay interlocking for the three team-selection paths. The screenshot does not visibly show X3 reset or Y3 buzzer logic, so these functions must be added to complete the written problem statement.



5.2 Future Scope

1. Multi-Hopper Carousel Expansion: Expanding the PLC I/O array to control 4 to 6 independent medicine hoppers simultaneously using indexed addressing.
2. Biometric Patient Authentication: Adding a fingerprint or facial recognition sensor on the dispenser front panel to prevent unauthorized access to controlled medications.
3. Voice Assistant Integration: Linking Firebase functions with Google Assistant and Amazon

CHAPTER 6:CONCLUSION

The PLC-Based Priority Determination System demonstrates the use of discrete inputs, outputs, internal relays and interlocking logic to identify the first team to press a push button. The selected team indicator gives visual feedback, while the buzzer provides an audible alert. The host reset prepares the system for the next question. The supplied GX Works3 screenshot provides a starting point for the three team-selection paths. To satisfy the complete Exercise 4.2 statement, the finished program must also include the host reset input X3 and buzzer output Y3, and these functions must be verified by simulation or hardware testing

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